Sahaj Bhadja

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Summary

Software engineer with 2+ years building **backend applications**, **REST APIs**, and **data processing systems**. Strong foundation in Python, Java, SQL databases, and software engineering best practices including automated testing and cross-functional collaboration.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Data Structures

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Algorithms

Experience

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Designed and developed **backend services using Python/Flask**, implementing REST APIs with JSON data formats that handled 10K+ daily requests with comprehensive error handling and logging.
- Wrote **clean, maintainable, and testable code** with clear documentation, participating in code reviews and collaborating with cross-functional teams to implement new customer requirements and features.

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **backend applications with SQL databases** (PostgreSQL) serving 500+ users, designing APIs and optimizing queries that reduced response latency from **1.2s to 350ms** through indexing and caching.
- Developed **data processing workflows** handling 5K+ monthly jobs, implementing validation logic and automated error handling while maintaining existing codebase through investigation and resolution of problem areas.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **data processing pipelines using Python** to handle 100K+ records, implementing algorithms for feature extraction and transformation with comprehensive logging and documentation.
- Collaborated with **cross-functional research teams** to design and develop software solutions, demonstrating clear documentation through flowcharts and well-structured code following engineering best practices.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **REST APIs and data pipelines** using Python and AWS services, implementing event-driven patterns that improved processing efficiency by **68%** with comprehensive automated testing.
- Maintained existing codebase by **investigating and resolving problem areas**, writing reusable code components and adhering to modern engineering standards in Agile/Scrum sprints with regular code reviews.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Developed **backend services with RDBMS** (Azure SQL, PostgreSQL) and NoSQL databases (MongoDB), implementing SQL queries and data workflows for 8 microservices handling 2M+ records.
- Built **REST APIs using JSON/XML formats** for client integrations, collaborating with cross-functional teams to troubleshoot and debug software programs across distributed systems.

Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL
Backend & APIs: REST APIs, JSON, XML, Flask, FastAPI, Spring Boot basics, event-driven patterns
Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL queries, data workflows, indexing
Tools & Practices: Git, Docker, AWS, Azure, CI/CD, automated testing, Agile/Scrum, code reviews

Projects

HMIS Code | *Python, FastAPI, PostgreSQL, React, Docker*

- Built a **full-stack application with REST APIs** and PostgreSQL backend, implementing clean architecture with reusable components, comprehensive documentation, and automated testing for reliability.

Linguistix Code | *Python, AWS (ECS, Fargate), Docker, CI/CD*

- Developed a **Python-based application** with automated deployment pipelines, implementing validation gates and engineering best practices for high-quality, maintainable code.

Big-Brain Live | *Python, PostgreSQL, REST APIs, GCP*

- Built a **backend system with SQL database** and REST APIs, implementing data processing workflows with clear documentation and well-structured code for maintainability.